

4. Linear Regression and Randomized Experiments

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Roadmap

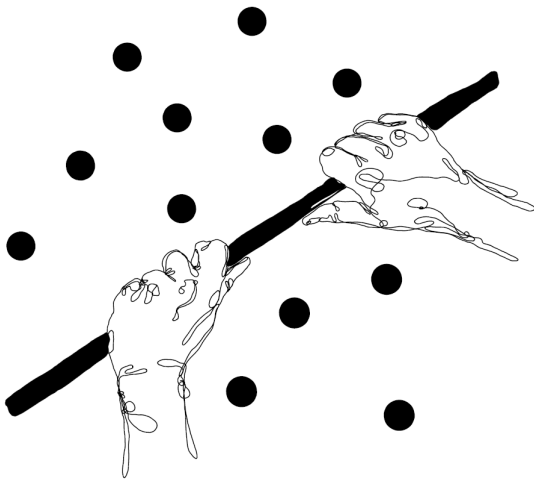
1. Regression with no covariates
2. Linear regression with covariates
3. Regression for stratified experiments
4. Cluster randomized experiments

Where are we? Where are we going?

- So far: analysis of experiments with Fisher's and Neyman's approaches.
 - Neyman: Unbiased estimators, (conservative) variances.
 - Fisher: exact test of the sharp null.
- Today: how does the workhorse estimator, OLS, fit into this story?

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- So far: analysis of experiments with Fisher's and Neyman's approaches.
 - Neyman: Unbiased estimators, (conservative) variances.
 - Fisher: exact test of the sharp null.
- Today: how does the workhorse estimator, OLS, fit into this story?
- Why would we consider using regression?
 - **Simplicity:** known tool that is already very common.
 - **Increased precision:** we may want to add covariates for more precise effect estimates.



Source: *Chapter 13 of The Effect (Textbook 2)* by Nick Huntington-Klein

1/ Regression with no covariates

Analysing Experiments with Regression?

- Q: Under complete randomization, can we use OLS to estimate ATEs
 - Literally, just $\text{lm}(y \sim d)$?
- Recall that the OLS estimator solves the least squares problem:

$$(\hat{\tau}_{\text{ols}}, \hat{\alpha}_{\text{ols}}) = \arg \min_{\tau, \alpha} \sum_{i=1}^n (Y_i - \alpha - \tau D_i)^2 \quad (1)$$

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- The coefficient on a binary r.v. is mechanically the diff. in means:

$$\hat{\tau}_{\text{ols}} = \bar{Y}_1 - \bar{Y}_0 = \hat{\tau}_{\text{diff}} \quad (2)$$

- Standard Neyman analysis for unbiasedness, sampling variance.
- Generalized to discrete treatments with > 2 levels.

Justifying the Linear Model

- Mechanically the same, but can we justify the linear model itself?
 - Key assumptions: **linearity** and **mean independence of errors**.
- Some simple manipulation of the consistency assumption:

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- “Linear” functional form fully justified by consistency alone with:
 - Intercept $\alpha = \mathbb{E}[Y_i(0)]$ is the average control outcome.
 - Slope $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$ is the PATE.
 - Error is deviation for control PO + treatment effect heterogeneity.

Mean Independent Errors

$$\varepsilon_i = (Y_i(0) - \mathbb{E}[Y_i(0)]) + D_i \cdot (\tau_i - \tau)$$

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- Randomization + consistency $\rightsquigarrow$ linear model.
 - Does not imply homoskedasticity or normal errors, though!

Homoskedasticity

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 - True under constant treatment effects!
- Under homoskedasticity, variance of the OLS estimator is:

$$\mathbb{V}[\widehat{\tau}_{\text{ols}} \mid \mathbf{D}] = \frac{\sigma^2}{\sum_{i=1}^n (D_i - \overline{D})^2}$$

Variance Estimation

- “Standard” variance estimator under homoskedasticity:

$$\widehat{V}_{\text{const}} = \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{\varepsilon}_i^2}{\sum_{i=1}^n (D_i - \bar{D})^2} = \frac{\frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\alpha}_{\text{ols}} - \hat{\tau}_{\text{ols}} D_i)^2}{\sum_{i=1}^n (D_i - \bar{D})^2}$$

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- We can rewrite this as a function of the **pooled** variance $\widehat{\sigma}_{Y|D}^2$:

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$$\widehat{\sigma}_{Y|D}^2 = \frac{1}{n-2} \left(\sum_{i:D_i=0} (Y_i - \overline{Y}_0)^2 + \sum_{i:D_i=1} (Y_i - \overline{Y}_1)^2 \right)$$

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- **Inconsistent:** $\widehat{V}_{\text{const}} - \mathbb{V}[\widehat{\tau}] \xrightarrow{p} c \neq 0$ unless
 - Homoskedasticity holds: $\sigma_1^2 = \sigma_0^2$
 - Design is balanced: $n_1 = n_0$

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 - Bias:

$$\begin{aligned} & \mathbb{E}(\widehat{\mathbb{V}}_{\text{const}}) - \mathbb{V}[\widehat{\tau}] \\ &= \underbrace{\mathbb{E}\left(\frac{\frac{1}{n-2} \sum_{i=1}^n \widehat{\varepsilon}_i^2}{\sum_{i=1}^n (D_i - \overline{D})^2}\right)}_{\text{under const. effect assumption}} - \underbrace{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}\right)}_{\text{true variance}} \\ &= \frac{(n_1 - n_0)(n - 1)}{n_1 n_0 (n - 2)} (\sigma_1^2 - \sigma_0^2) \neq 0 \end{aligned}$$

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- Unless:

- Homoskedasticity holds: $\sigma_1^2 = \sigma_0^2$
 - Constant effect: $Y_i(1) - Y_i(0) = \text{const.}$
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Robust SEs

- Eicker-Huber-White (EHW) robust/sandwich variance estimator:

$$\underbrace{\widehat{\mathbf{V}}_{\text{EHW}}}_{\text{sandwich}} = \underbrace{\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1}}_{\text{bread}} \underbrace{\left(\sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i' \right)}_{\text{meat}} \underbrace{\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \right)^{-1}}_{\text{bread}}$$
$$= (\mathbb{X}'\mathbb{X})^{-1} \left(\sum_{i=1}^n \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbb{X}'\mathbb{X})^{-1} \quad \text{where } \mathbb{X} = [\mathbf{1} \quad \mathbf{D}]$$

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- Recall the PATE-targeted variance of the difference-in-means:

$$\mathbb{V}(\widehat{\tau}_{\text{diff}}) = \frac{\sigma_0^2}{n_0} + \frac{\sigma_1^2}{n_1} = \frac{\mathbb{V}[Y_i(0)]}{n_0} + \frac{\mathbb{V}[Y_i(1)]}{n_1}$$

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- To see this, we can derive $\widehat{\mathbf{V}}_{\text{EHW}}$ under our case:

$$\widehat{\mathbf{V}}_{\text{EHW}} = \frac{\widetilde{\sigma}_1^2}{n_1} + \frac{\widetilde{\sigma}_0^2}{n_0}, \quad \text{where } \widetilde{\sigma}_d^2 = \frac{1}{n_d} \sum_{i:D_i=d} (Y_i - \bar{Y}_d)^2$$

- $\widetilde{\sigma}_0^2, \widetilde{\sigma}_1^2$ consistent for $\sigma_0^2, \sigma_1^2 \rightsquigarrow \widehat{\mathbf{V}}_{\text{EHW}}$ consistent for $\mathbb{V}(\widehat{\tau}_{\text{diff}})$

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- Leverage: $h_{ii} = \mathbf{x}_i (\mathbb{X}'\mathbb{X})^{-1} \mathbf{x}_i'$
- In this setting: $h_{ii} = n_1^{-1}$ if $D_i = 1$ and $h_{ii} = n_0^{-1}$ if $D_i = 0$

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- Samii & Aronow (2012): HC2 is exactly the Neyman variance estimator:

$$\widehat{\mathbb{V}}_{\text{HC2}} = \frac{\widehat{\sigma}_0^2}{n_0} + \frac{\widehat{\sigma}_1^2}{n_1}$$

- $\rightsquigarrow$ simple OLS + HC2 = unbiased point and variance estimator.

Application in R Using (Sim) AI Exp. Data

```
1 > AI_data <- as_tibble(read.csv(url("https://bit.ly/3FHsusw"))); AI_data
2 # A tibble: 500 x 8
3   treat_ind test_outcome_pre test_outcome_post student_age student_gender tutor_age
4   <int>      <int>          <int>          <int>      <int>      <int>
5 1         1          0            1           11         0         27
6 2         1          0            1            6         1         45
7 3         0          1            1           15         1         54
8 4         0          1            1            6         1         50
9 5         0          1            0            8         1         64
10 6         1          1            1           15         1         43
11 7         1          0            1            7         1         47
12 8         1          1            1           16         0         49
13 9         0          0            1           12         1         42
14 10        1          1            1           11         1         28
15 # ... 490 more rows
16 # ... 2 more variables: years_of_experience <int>, education_level <chr>
17 # Use `print(n = ...)` to see more rows
18
19 > lm1 <- lm(test_outcome_post ~ treat_ind, data = AI_data)
20 > vcovM <- sandwich::vcovHC(lm1, type = 'HC2')
21 > sqrt(vcovM[1,1])
22 [1] 0.02955412
23 > sqrt(vcovM[2,2]) # sqrt(diag(vcovM))
24 [1] 0.04209018
25
26 > # Or
27 > estimatr::lm_robust(test_outcome_post ~ treat_ind, AI_data, se_type = 'HC2')
28       Estimate Std. Error  t value    Pr(>|t|)    CI Lower CI Upper DF
29 (Intercept) 0.65134100 0.02955412 22.038927 1.233662e-75 0.59327487 0.7094071 498
30 treat_ind    0.03903557 0.04209018  0.927427 3.541541e-01 -0.04366065 0.1217318 498
```

2/ Linear regression with covariates

Adding Covariates

- What if we add covariates to our regression model?

$$(\hat{\tau}_{\text{adj}}, \hat{\alpha}_{\text{adj}}, \hat{\beta}_{\text{adj}}) = \arg \min_{\tau, \alpha, \beta} \sum_{i=1}^n (Y_i - \alpha - \tau D_i - \tilde{\mathbf{x}}_i' \beta)^2$$

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- Why might we do this? To increase **precision** of our estimates.
 - We hope $\mathbb{V}[\hat{\tau}_{\text{adj}}] < \mathbb{V}[\hat{\tau}_{\text{diff}}]$ so we have smaller CIs, more powerful tests.
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 - Intuition: less residual variation in Y_i after accounting for $\mathbf{X}_i$
- Questions:
 - Is $\hat{\tau}$ still unbiased? Consistent?
 - Should we expect an increase in precision?
 - Controversial! Freedman (2008) “Randomization does not justify the regression model”

OLS is biased, but consistent (Freedman, 2008. Adv. in Appl. Math)

- Agnostic approach: don't assume correctness of the linear model.
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- Freedman (2008) shows the same thing for finite-sample inference.

Variance of Adjustment Estimator

- Complete randomization + single, mean-zero covariate X_i
 - Generalizes easily to more covariates.
 - Let $\sigma_{0x} = \text{cov}(Y_i(0), X_i)$ and $\sigma_{1x} = \text{cov}(Y_i(1), X_i)$.
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- Freedman (2008) derived gains from adjusting for X_i using OLS:

$$\mathbb{V}[\widehat{\tau}_{\text{diff}}] - \mathbb{V}[\widehat{\tau}_{\text{adj}}] = \frac{\sigma_{0x} \{ \sigma_{0x} + 2(1 - 2p)\sigma_{1x} \}}{np(1 - p)}$$

Variance of Adjustment Estimator

- Complete randomization + single, mean-zero covariate X_i
 - Generalizes easily to more covariates.
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- Will adjustment decrease the sampling variance?
 - If design is balanced, $p = 1/2$, then adjustment always helps.
 - Design imbalance could lead to adjustment hurting.
- Estimation: EHW robust variance estimators are consistent or asymptotically conservative for $\mathbb{V}[\widehat{\tau}_{\text{adj}}]$

Regression with Full Interactions

- OLS estimator from fully interacted model, $\widehat{\tau}_{\text{inter}}$:

$$Y_i = \alpha + \tau D_i + \widetilde{\mathbf{X}}_i' \beta + D_i \widetilde{\mathbf{X}}_i' \gamma + \varepsilon_i$$

- Equivalent to running separate Y_i on $\widetilde{\mathbf{X}}_i$ in each D_i
- As with non-interacted model, $\widehat{\tau}_{\text{inter}}$ is consistent for τ and asymptotically normal.
- Lin (2013): **fully interacted model will never hurt precision asymptotically.**
 - Freedman critique was right, but Lin shows an easy way to resolve.
- EHW robust variance estimators are consistent or asymptotically conservative.

Linear Regression with Covariates

In R

```
# Step 1: Compute centered covariates
your_data$Xtilde <- NULL

# Step 2: Write down your formula
your_formula <- NULL

# Step 3: Fit the model using lm() or estimatr::lm_robust()
your_fitted_model <- lm(your_formula, data = your_data)

# Step 4: Compute robust standard errors (skip if you used lm_robust)
your_vcov <- sandwich::vcovHC(your_fitted_model, type = 'HC2')

# Step 5: Check the point and se estimate of your coefficients
# (look for tau hat!)
est <- cbind("coef" = your_fitted_model$coef,
            "se" = sqrt(diag(your_vcov)))
```

Example Code

```
1 > AI_data <- AI_data |>
2   mutate(Xtilde = student_age - mean(student_age)) |>
3   select(treat_ind, test_outcome_post, Xtilde); head(AI_data,3)
4 # A tibble: 3 x 3
5   treat_ind test_outcome_post Xtilde
6     <int>         <int>    <dbl>
7 1         1             1 -0.566
8 2         1             1 -5.57
9 3         0             1  3.43
10
11 > estimatr::lm_robust(test_outcome_post ~ treat_ind * Xtilde, data = AI_data)
12               Estimate Std. Error   t value    Pr(>|t|)    CI Lower    CI Upper   DF
13 (Intercept)    0.649860304 0.029645673 21.9209157 5.557487e-75 0.591613722 0.708106886 496
14 treat_ind      0.038364943 0.042245496  0.9081428 3.642438e-01 -0.044637245 0.121367131 496
15 Xtilde         0.008351571 0.008581795  0.9731730 3.309417e-01 -0.008509581 0.025212723 496
16 treat_ind:Xtilde -0.019462905 0.012003266 -1.6214674 1.055529e-01 -0.043046422 0.004120612 496
17
18 > your_fitted_model <- lm(test_outcome_post ~ treat_ind * Xtilde, data = AI_data)
19 > vcovM_adj <- sandwich::vcovHC(your_fitted_model, type = 'HC2'); sqrt(diag(vcovM_adj))
20               (Intercept)      treat_ind      Xtilde treat_ind:Xtilde
21               0.029645673      0.042245496      0.008581795      0.012003266
22
23 > est <- cbind("coef" = your_fitted_model$coef,
24               "se" = sqrt(diag(your_vcov))); est
25               coef      se
26 (Intercept)    0.649860304 0.029645673
27 treat_ind      0.038364943 0.042245496
28 Xtilde         0.008351571 0.008581795
29 treat_ind:Xtilde -0.019462905 0.012003266
```


Summarizing Regression

- Regression with no covariates = standard Neyman analysis.
- Regression with (uninteracted) covariates:
 - Consistent for SATE/PATE.
 - Usually will help with precision, but can hurt.
- Regression with interacted covariates:
 - Consistent for SATE/PATE.
 - Asymptotically will never hurt precision.
- Always use robust/HC2 variance estimators unless you have good reasons.

3/ Regression for stratified experiments

Cf. Imbens & Rubin (2015), *Causal Inference for Statistics, Social, and Biomedical Sciences*, Ch. 9.6

Regression for Stratified Experiments

- Setup: block randomized experiment with block indicators W_{ij} .
 - Block “fixed effects” $W_{ij} = 1$ if i is in block j , 0 otherwise.
 - Blocks $j \in \{1, \dots, J\}$ with sizes $w_j = n_j/n$ and propensity scores $p_j = n_{1,j}/n_j$.
- STAR Project (Krueger, 1999): within each school (block), classes were randomized.

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- Naive approach: just include the block FEs in OLS

$$(\hat{\tau}_{b,fe}, \hat{\alpha}_1, \dots, \hat{\alpha}_J) = \arg \min_{(\tau, \alpha_1, \dots, \alpha_J)} \sum_{i=1}^n \left(Y_i - \tau D_i - \sum_{j=1}^J \alpha_j W_{ij} \right)^2$$

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- $\hat{\tau}_{b,fe}$ **not consistent** for the PATE unless:

$$\hat{\tau}_{b,fe} \xrightarrow{p} \frac{\sum_{j=1}^J \omega_j \tau_j}{\sum_{j=1}^J \omega_j}, \quad \text{where} \quad \omega_j = w_j p_j (1 - p_j)$$

- Propensity scores are equal across blocks: $p_j = p$ for all j .
- ATEs are equal across strata $\tau_j = \tau$ for all j .

Block Randomized Trials: Correct Analysis

1. Just use original Neyman analysis aggregating within-strata analyses.
 2. Weight OLS by inverse of the propensity score: $1/p_j$.
 3. Fully interact block FEs with treatment.
- Latter two allow for additional covariates to be added.
 - Check Imbens and Rubin (2015) Ch.9.6.1., second model
 - See this simulation study using DeclareDesign:
<https://declaredesign.org/blog/posts/biased-fixed-effects.html>

Block Randomized Trials: Correct Analysis

2. Weight OLS by inverse of the propensity score.

$$(\hat{\tau}_{b,fe}, \hat{\alpha}_1, \dots, \hat{\alpha}_J) = \arg \min_{(\tau, \alpha_1, \dots, \alpha_J)} \sum_{i=1}^n s_i \left(Y_i - \tau D_i - \sum_{j=1}^J \alpha_j W_{ij} \right)^2$$

$$\text{where } s_i = \sum_{j=1}^J \left\{ \left(\frac{1}{p_j} \right) D_i + \left(\frac{1}{1 - p_j} \right) (1 - D_i) \right\} W_{ij} \quad \text{and} \quad p_j = \frac{n_{1j}}{n_j}.$$

In R

```
your_formula <- as.formula("outcome ~ treat + x_tilde1 + x_tilde2")

your_data <- data.frame(outcome, treat,
                        x_tilde1, x_tilde2,
                        weights, block)

your_fitted_model <- estimatr::lm_robust(your_formula, data = your_data,
                                         weights = weights, # s
                                         se_type = "HC2",
                                         fixed_effects = block)
```

4/ Cluster randomized experiments

Cf. Angrist & Pischke (2008), *Mostly Harmless Econometrics*, Ch. 8

Clustering Treatments

- Treatment often allocated at a higher level than the data.
 - Counties are treated, but we have individual-level data.
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 - Classrooms are treated, but we have student data.
- Has considerable benefits:
 - Often cheaper/easier to implement than individual assignment.
 - Allows for interference within clusters without bias.
- But lots of confusion about how to analyze.
 - More valuable to add more individuals or clusters?
 - What to do with individual-level covariates?

Cluster Randomized Trials

- Setup:
 - Clusters: $k \in \{1, \dots, K\}$
 - Randomly choose K_1 treatment clusters, K_0 control.
 - Each cluster has units $i \in \{1, \dots, m_k\}$ with $\sum_{k=1}^K m_k = n$
 - Treatment assignment at cluster level: $D_{ik} = D_k$
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 - Potential outcomes $Y_{ik}(d)$
- Random assignment at the cluster level: $\{Y_{ik}(1), Y_{ik}(0)\} \perp\!\!\!\perp D_j$.
- Quantity of interest still at individual level:

$$\text{SATE} = \frac{1}{n} \sum_{k=1}^K \sum_{i=1}^{m_k} \{Y_{ik}(1) - Y_{ik}(0)\}$$

Analysis of Clustered Experiments

- Simple setting: all clusters have the same size $m_k = m$ for all k .
- Simple difference in means is unbiased:

$$\begin{aligned}\hat{\tau}_{\text{cl}} &= \frac{1}{mK_1} \sum_{k=1}^K \sum_{i=1}^{m_k} D_k Y_{ik} - \frac{1}{mK_0} \sum_{k=1}^K \sum_{i=1}^{m_k} (1 - D_k) Y_{ik} \\ &= \frac{1}{K_1} \sum_{k=1}^K D_k \bar{Y}_k - \frac{1}{K_0} \sum_{k=1}^K (1 - D_k) \bar{Y}_k\end{aligned}$$

- $\bar{Y}_k$ is the cluster average: $\frac{1}{m} \sum_{i=1}^m Y_{ik}$
- Unbiasedness follows from Neyman-style analysis at cluster level.
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 - Unbiasedness follows from Neyman-style analysis at cluster level.
 - Estimator is biased, but consistent (in K) if cluster size varies.
- Neyman-style conservative variance:

$$\mathbb{V}[\hat{\tau}_{\text{cl}} \mid \mathbf{D}] \leq \frac{\mathbb{V}[\bar{Y}_k(1)]}{K_1} + \frac{\mathbb{V}[\bar{Y}_k(0)]}{K_0} \quad \text{where } \bar{Y}_k(d) = \frac{1}{m} \sum_{i=1}^m Y_{ik}(d)$$

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- $\mathbb{X}' = [1 \quad \mathbf{D}]'$, $\mathbb{X}'_k = [1 \quad \mathbf{D}_k]'$, and $\widehat{\boldsymbol{\varepsilon}}_k = (\widehat{\varepsilon}_{1k}, \dots, \widehat{\varepsilon}_{m_k k})$
- Consistent as the number of clusters grows.

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 - Consistent as the number of clusters grows.
- **Cluster at the treatment assignment level** (no higher or lower).
- Vanilla CRVE is biased, Bell & McCaffrey proposed CR2 adjustment similar to HC2 (usually preferable).

Cluster Randomized Trials: Analysis

In R

```
your_formula <- as.formula("outcome ~ treat + x_tilde1 + x_tilde2")

your_data <- data.frame(outcome, treat,
                        x_tilde1, x_tilde2,
                        cluster)

your_fitted_model <- estimatr::lm_robust(your_formula, data = your_data,
                                         clusters = cluster,
                                         se_type = "CR2")

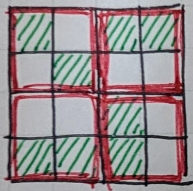
??estimatr::lm_robust # Check more options for se_type

# Or
your_model <- lm(your_formula, data = your_data)
your_vcov <- clubSandwich::vcovCR(your_model, cluster = your_data$cluster,
                                  type = "CR2")
```

- You may have block and cluster design at the same time.

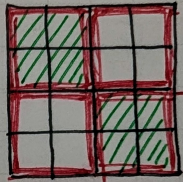
/// : treated

□ : block



/// : treated

□ : cluster



Onto the presentations & discussions!

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Appendix

Cost of Clustering

- Standard variance under **individual assignment**:

$$\mathbb{V}[\hat{\tau}_{\text{diff}}] = \frac{\mathbb{V}[Y_{ik}(1)]}{mK_1} + \frac{\mathbb{V}[Y_{ik}(0)]}{mK_0}$$

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- Under clustering, the variance of the cluster average inflates:

$$\frac{\mathbb{V}[\bar{Y}_k(1)]}{K_1} = \frac{\mathbb{V}[Y_{ik}(1)]}{mK_1} (1 + (m-1)\rho_1)$$

- Intraclass correlation coefficient** (ICC):

$$\rho_1 = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j \neq i} \mathbb{E} \left\{ (Y_{ik}(1) - \bar{Y}(1))(Y_{jk}(1) - \bar{Y}(1)) \right\}$$

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- ρ measures how similar units are within clusters.
 - Usually cluster is less efficient because $\rho > 0$.
 - More similarity $\rightsquigarrow$ each unit provides redundant information $\rightsquigarrow$ less efficiency under clustering.

Linear Regression and Causality

- Regression: **conditional expectation function** of Y given $\mathbf{X}$

$$\mathbb{E}(Y|\mathbf{X}) = f(\mathbf{X}) = \beta^T \mathbf{X}$$

- Q: When can we interpret coefficients as causal effects?
- Causal model as structural equation model:

$$Y_i(d) = \alpha + \tau d + \varepsilon_i \quad \text{for } d = 0, 1, \text{ where } \mathbb{E}(\varepsilon_i) = 0$$

1. No interference between units
 2. $\mathbb{E}(Y_i(0)) = \alpha$
 3. $Y_i(1) - Y_i(0) = \tau$ for all $i \rightsquigarrow$ **constant unit-level causal effect**
- Heterogeneous treatment effect model:

$$Y_i(d) = \alpha + \tau_i d + \varepsilon_i = \alpha + \tau d + \underbrace{(\tau_i - \tau)d}_{=\varepsilon_i(d)} + \varepsilon_i$$

where $\mathbb{E}(\varepsilon_i) = 0$ and $\tau = \mathbb{E}(\tau_i) = \mathbb{E}(Y_i(1) - Y_i(0))$

- $\mathbb{E}(\varepsilon_i(d)) = 0$ for $d = 0, 1$
- $\alpha = \mathbb{E}(Y_i(0))$